Software Architecture

AST Solutions

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AST Solutions

PROBLEM

Nexus Safe-Cloud currently manages its information using spreadsheets, documents, emails, and separate tools. This creates duplicate or outdated information and makes communication between areas difficult.

This causes repeated work, errors in management and billing, little visibility into customers and projects, and possible security risks, especially when working with remote employees.

Therefore, the company needs a secure technology solution that centralizes information, connects the different areas, and improves operations and visibility while maintaining security, availability, performance, and scalability.

Architecture Drivers

High Priority



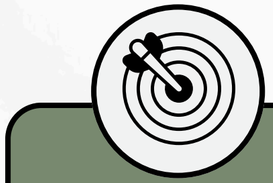
Availability and Resilience

Continuous access from offices and remote locations.



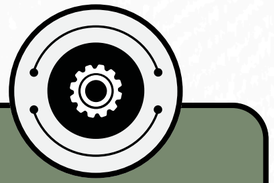
Security

Protected access through ZTNA + MFA.



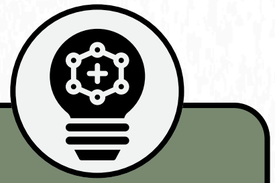
Data Consistency

Centralized information without duplicates.



Performance

Almost real-time responses.



Scalability

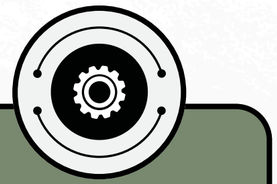
Support the growth of users and data.



Interoperability

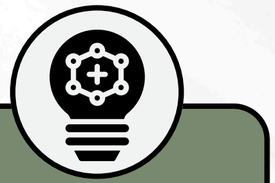
Secure and redundant connectivity between offices.

Medium Priority



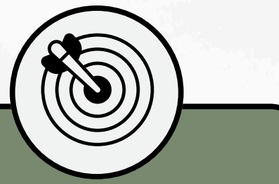
Traceability and Auditing

Track changes and transactions.



Usability

Make ERP adoption easier.



Maintainability and Evolvability

Add new modules without affecting existing ones.

Requirements

FUNCTIONAL

MODULE	FEATURES
Management	Management dashboard - PDF/Excel reports
Customers	Single profile - Contract/project history - Business opportunities
Projects	Stage and task management - Owners and dates - Progress and hours tracking - Risk alerts
Billing	Automatic and electronic billing - Unified accounts receivable - Due-date control - Change auditing
Resources - Teams	Availability and assignment - Over-assignment prevention - ZTNA Management · Access policies by role/user - Access auditing

NON-FUNCTIONAL

ATTRIBUTE	FEATURES
Security	SSO + MFA - Role-based authorization - Information encryption
Performance	Responses in less than 2 seconds - Dashboards updated at least every 5 minutes - Scalability
Availability	Access from both offices and remotely - Independent modules
Usability	Access from mobile devices/laptops - No technical knowledge required

Domain-Oriented Hybrid Architecture

ERP organized by domains: Customers, Projects, Billing, Resources, and Reporting.
Progressive evolution through Strangler Fig, starting with a modular core and extracting services as needed.
Reduces risks and avoids a direct migration.

Architectural Patterns

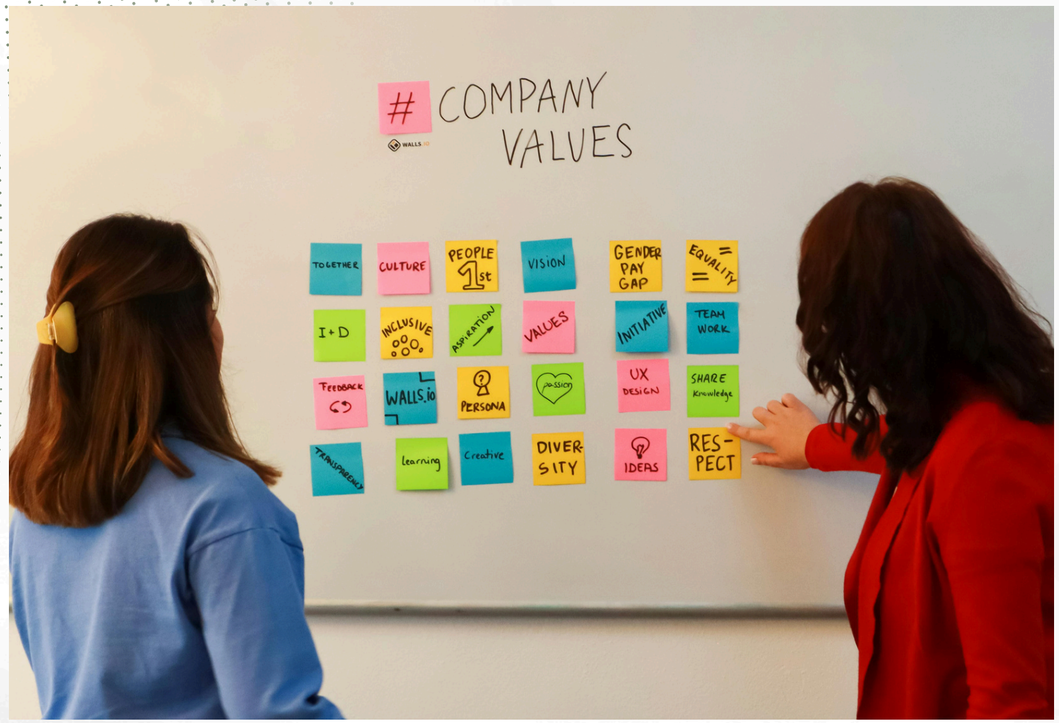
API Gateway + BFF: Centralized and adapted access.
Event-Driven + Pub/Sub: Communication and synchronization between modules.
CQRS: Query and dashboard optimization.
Circuit Breaker / Retry / Bulkhead: Resilience against failures.

Integration

Message Broker for asynchronous communication.
REST/GraphQL APIs for external systems.
ETL for data migration, cleaning, and deduplication.
Webhooks for event notifications.

Network and Security

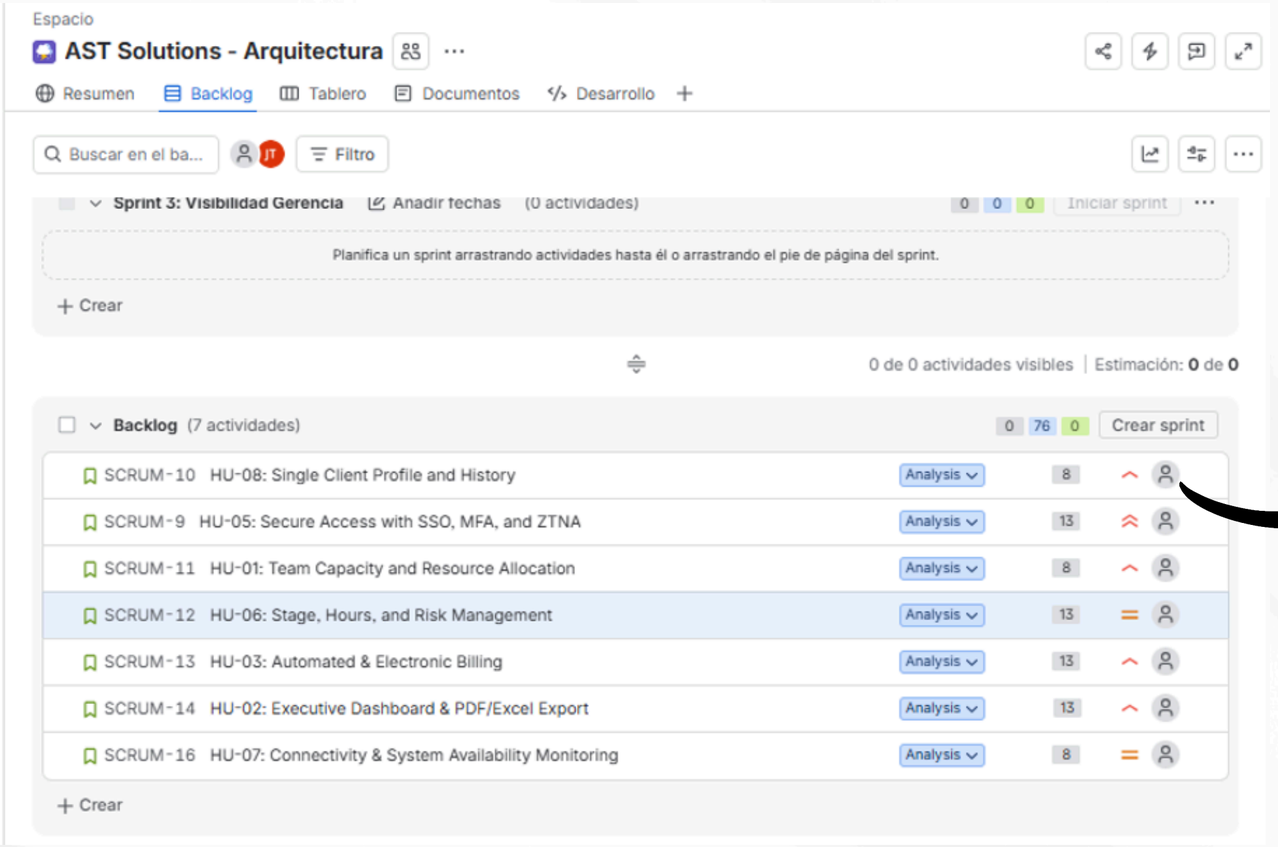
SD-WAN between Medellín, Bogotá, and the cloud, with failover.
Hybrid cloud with multi-zone deployment.
ZTNA + MFA + SSO + RBAC/ABAC for secure access.
Monitoring and SIEM/NOC for traceability and auditing.

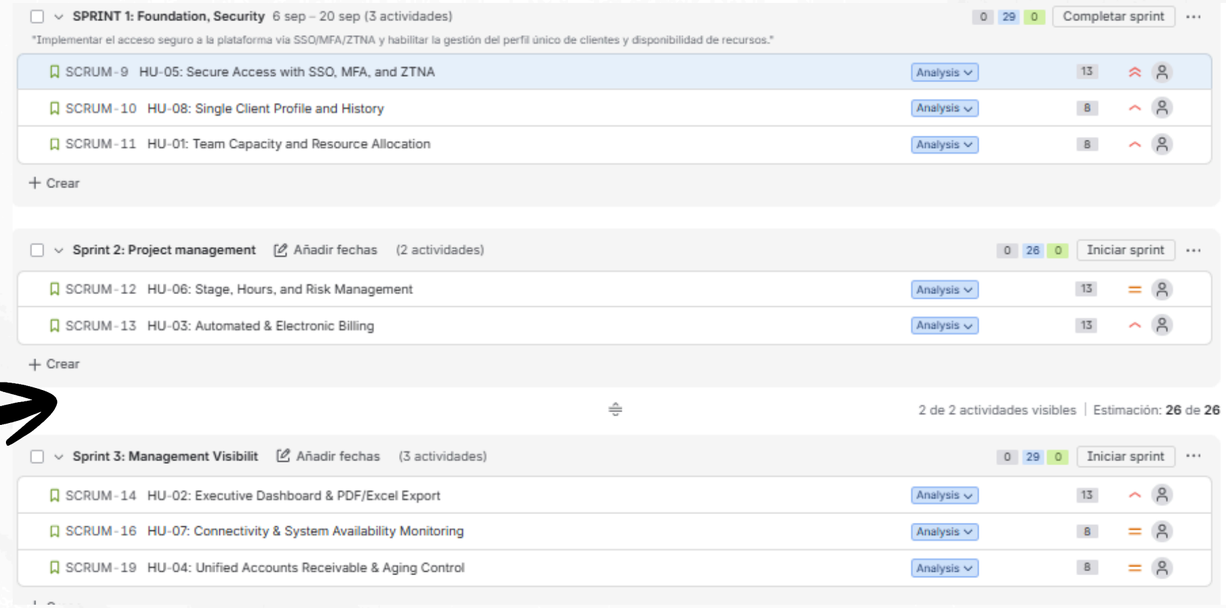


Value Proposition

- **Predictive capability: We anticipate over-assignment and capacity risks before they happen.**
- **ZTNA as a business advantage: We turn ZTNA security into an element that Nexus can use to strengthen its offer to customers.**
- **Guided migration: We manage the gradual transition of data and processes through Strangler Fig + Kanban, reducing risks for Nexus.**

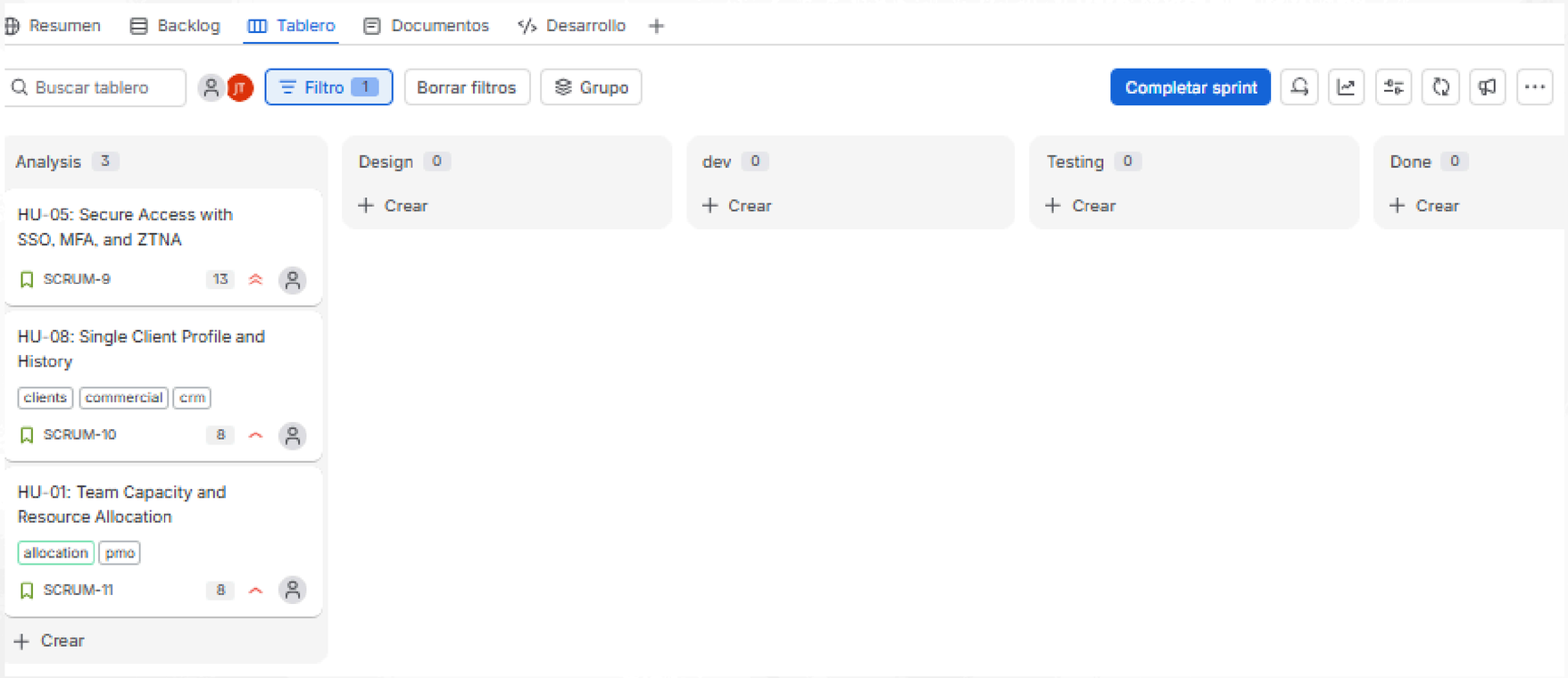
Kanban-Scrum





Board Workflow

The Sprint Board shows the team's work flow using custom Kanban states (Analysis - Design - Development - Testing - Done). Items in progress allow real-time tracking of workload distribution



The ERP will centralize information and remove isolated information between offices, processes, and remote employees.
The proposed architecture prioritizes security, availability, consistency, performance, and scalability.
The evolutionary and modular approach supports the system’s gradual growth and reduces the risks of a complete migration.
Patterns such as Event-Driven, API Gateway, CQRS, Saga, and ZTNA provide an interoperable and sustainable solution.
The requirements and Kanban connect the architecture with development, making it easier to deliver value progressively.



Conclusion

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THANK YOU

Team 3